

Labolatoria nr 5

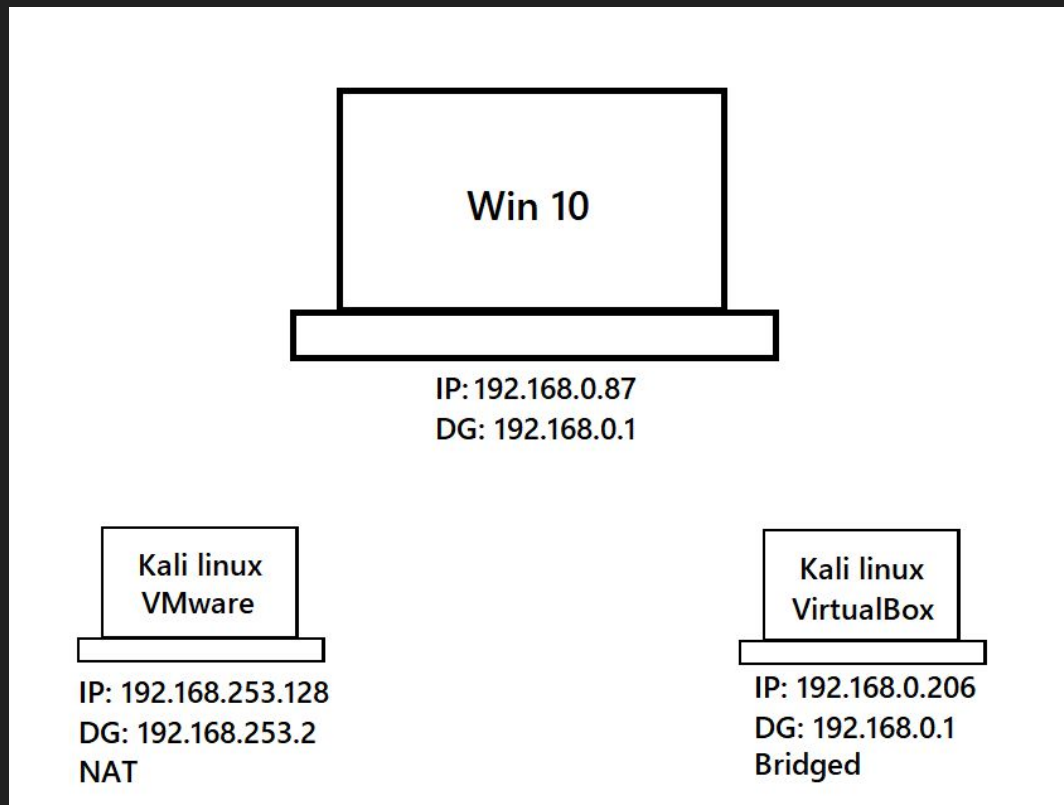
Bartłomiej Nałódka

Zad 1. Przygotowanie środowiska

Na swoim komputerze z Windowsem 10 uruchomiłem dwie maszyny wirtualne Kali Linux. Jedną w trybie Bridged, a drugą w trybie NAT.

Urządzenia są w stanie pingować się parami.

Maska dla każdej z podsieci to: 255.255.255.0



Zad 2. VirtualBox

Widzimy informacje, które
podałem na slajdzie z
konfiguracją środowiska.

Są widoczne dwa interfejsy
sieciowe - lo do komunikacji w
obrębie tej jednej maszyny i
eth0 - interfejs do komunikacji
z innymi urządzeniami.

```
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.206 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::6da0:de3a:327c:694e prefixlen 64 scopeid 0x20<link>
    inet6 2a02:a31a:e08f:4600:ef84:ae83:98a7:9ee6 prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:ad:25:87 txqueuelen 1000 (Ethernet)
    RX packets 91 bytes 80248 (78.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 103 bytes 57689 (56.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
$ route -n
Kernel IP routing table

```

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Iface
0.0.0.0	192.168.0.1	0.0.0.0	UG	100	0	0	eth0
192.168.0.0	0.0.0.0	255.255.255.0	U	100	0	0	eth0

Skanujemy podsieć

Wykryto bramę domyślną jak i maszynę, z której przeprowadzaliśmy skan.

Natomiast maszyna z Windowsem nie została wykryta.

```
(kali㉿kali)-[~]  
$ nmap 192.168.0.0/24  
  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-12 07:33 EST  
Nmap scan report for 192.168.0.1  
Host is up (0.061s latency).  
Not shown: 997 closed tcp ports (conn-refused)  
PORT      STATE SERVICE  
53/tcp    open  domain  
80/tcp    open  http  
5000/tcp  open  upnp  
  
Nmap scan report for 192.168.0.206  
Host is up (0.000056s latency).  
All 1000 scanned ports on 192.168.0.206 are in ignored states.  
Not shown: 1000 closed tcp ports (conn-refused)  
  
Nmap done: 256 IP addresses (2 hosts up) scanned in 6.83 seconds  
  
(kali㉿kali)-[~]  
$
```

Skanujemy konkretny adres

Tym razem skanowanie przyniosło efekty. Widzimy kilka otwartych portów z usługami Microsoftu, oraz informacje o systemie operacyjnym: Windowsie.

Widać także, że na skanowanym porcie działa VMware (druga z naszych maszyn).

```
(kali㉿kali)-[~]
$ nmap -A 192.168.0.87
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-12 07:38 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.71 seconds

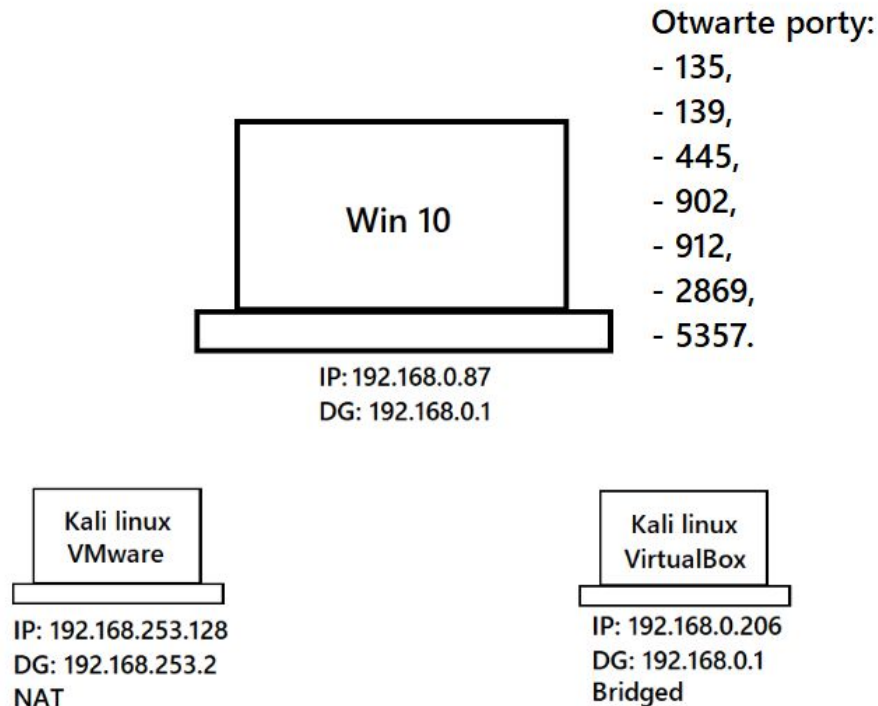
(kali㉿kali)-[~]
$ nmap -A -Pn 192.168.0.87
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-12 07:39 EST
Nmap scan report for 192.168.0.87
Host is up (0.00052s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT      STATE SERVICE          VERSION
135/tcp    open  msrpc            Microsoft Windows RPC
139/tcp    open  netbios-ssn      Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
902/tcp    open  ssl/vmware-auth  VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
912/tcp    open  vmware-auth      VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
2869/tcp   open  http             Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-title: Service Unavailable
|_http-server-header: Microsoft-HTTPAPI/2.0
5357/tcp   open  http             Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-title: Service Unavailable
|_http-server-header: Microsoft-HTTPAPI/2.0
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_nbstat: NetBIOS name: DESKTOP-5QFTIEH, NetBIOS user: <unknown>, NetBIOS MAC: 2c:f0:5d:ae:e5:fe (Micro-Star Intl)
|_smb2-security-mode:
|   3:1:1:
|_   Message signing enabled but not required
|_smb2-time:
|   date: 2024-12-12T12:39:29
|_   start_date: N/A

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 58.22 seconds
```

Rysunek techniczny wzbogacony o informacje pozyskane ze skanowania konkretnego adresu

Poznaliśmy numery otwartych portów działających na maszynie z Windowsem oraz usługi, jakie na nich działają.



Zad 2. VMware

Nie widać wielkiej różnicy
w porównaniu do VBox'a.

Można dodać, że flaga G
w odpowiedzi do polecenia
route -n wskazuje, że dany
adres jest bramą domyślną
dla danego hosta.

```
(kali㉿kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.253.128 netmask 255.255.255.0 broadcast 192.168.253.255
    inet6 fe80::a01:7c9a:2e9e:3fa0 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:16:9c:1c txqueuelen 1000 (Ethernet)
    RX packets 314 bytes 25695 (25.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 172 bytes 31547 (30.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 12 bytes 816 (816.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 12 bytes 816 (816.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali㉿kali)-[~]
$ route -n
Kernel IP routing table
Destination        Gateway           Genmask          Flags Metric Ref    Use Iface
0.0.0.0            192.168.253.2    0.0.0.0          UG    100    0      0 eth0
192.168.253.0      0.0.0.0          255.255.255.0    U     100    0      0 eth0
```


Skanujemy podsieci

Podobnie jak poprzednio, widzimy naszą maszynę i jej bramę domyślną.

Jedynym czynnym porcie na DG jest port nr 53 - standardowo używany do komunikacji z serwerami DNS.

```
(kali@kali)-[~]  
$ nmap 192.168.253.0/24  
  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-12 07:51 EST  
Nmap scan report for 192.168.253.2  
Host is up (0.00027s latency).  
Not shown: 999 closed tcp ports (conn-refused)  
PORT      STATE SERVICE  
53/tcp    open  domain  
  
Nmap scan report for 192.168.253.128  
Host is up (0.00025s latency).  
All 1000 scanned ports on 192.168.253.128 are in ignored states.  
Not shown: 1000 closed tcp ports (conn-refused)  
  
Nmap done: 256 IP addresses (2 hosts up) scanned in 2.52 seconds
```


Skanujemy konkretny adres

Wyniki bardzo podobne do poprzednich.

Przyczyną dla której -A nie działa, a -Pn już tak, jest to, że -Pn wyłącza sprawdzanie, czy host jest aktywny (czy odpowiada na icmp lub łączy się na porty 80. lub 443., co może być blokowane przez firewall).

```
(kali㉿kali)-[~]  
$ nmap -A 192.168.0.87
```

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-12 07:56 EST  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.61 seconds
```

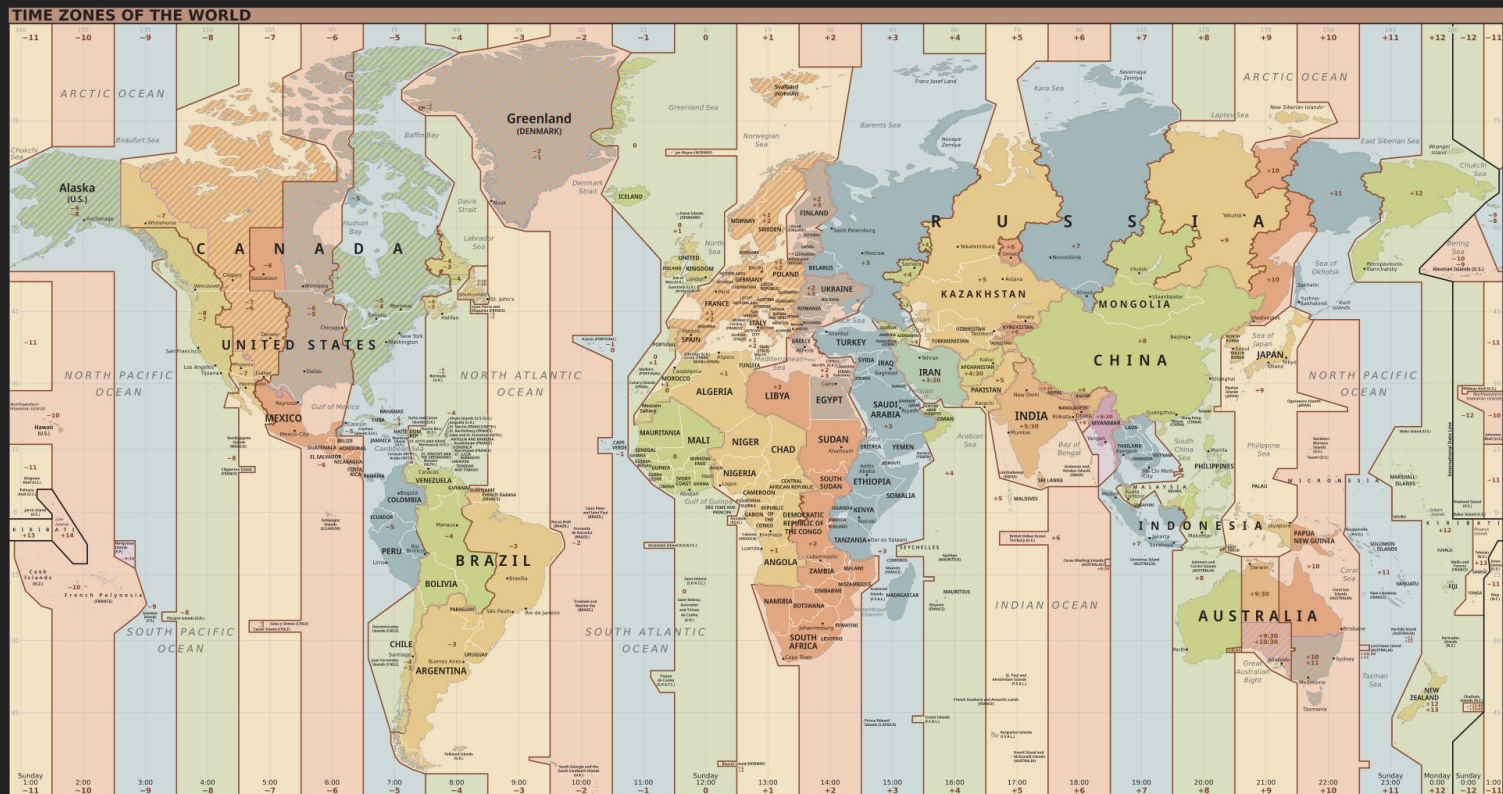
```
(kali㉿kali)-[~]  
$ nmap -A -Pn 192.168.0.87
```

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-12 07:56 EST  
Nmap scan report for 192.168.0.87  
Host is up (0.00078s latency).  
Not shown: 995 filtered tcp ports (no-response)  
PORT      STATE SERVICE          VERSION  
135/tcp   open  msrpc            Microsoft Windows RPC  
139/tcp   open  netbios-ssn     Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?  
912/tcp   open  vmware-auth     VMware Authentication Daemon 1.0 (Uses VNC, SOAP)  
5357/tcp  open  http            Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)  
|_http-server-header: Microsoft-HTTPAPI/2.0  
|_http-title: Service Unavailable  
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows
```

```
Host script results:  
| smb2-security-mode:  
|   3:1:1:  
|_   Message signing enabled but not required  
| smb2-time:  
|   date: 2024-12-12T12:56:59  
|_   start_date: N/A
```

```
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 35.30 seconds
```

Podczas skanowań uzyskaliśmy także informacje o czasie systemowym hosta, co może być pomocne w ustaleniu jego lokalizacji.



Zad 3. Ping z VMware na VBox

Pakiety wyglądają jak na obrazku. Zazaczyłem pakiety ICMP. Co ciekawe ich adresem źródłowym jest maszyna hosta - przyczyną tego może być fakt, że VMware ukrywa się za NATem.

Mamy tu jednak kilka innych informacji: czas przyścia pakietu, protokół, TTL, id oraz długość pakietu.

```
(kali@kali)-[~]
$ sudo tcpdump -i eth0 -v
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
08:22:37.134366 IP (tos 0x0, ttl 63, id 7480, offset 0, flags [none], proto ICMP (1), length 84)
    192.168.0.87 > 192.168.0.206: ICMP echo request, id 37027, seq 3, length 64
08:22:37.134387 IP (tos 0x0, ttl 64, id 24311, offset 0, flags [none], proto ICMP (1), length 84)
    192.168.0.206 > 192.168.0.87: ICMP echo reply, id 37027, seq 3, length 64
08:22:37.134738 IP (tos 0x0, ttl 64, id 43467, offset 0, flags [DF], proto UDP (17), length 72)
    192.168.0.206.53585 > compalhub.home.domain: 57174+ PTR? 206.0.168.192.in-addr.arpa. (44)
08:22:37.142475 IP (tos 0x0, ttl 64, id 10870, offset 0, flags [DF], proto UDP (17), length 72)
    compalhub.home.domain > 192.168.0.206.53585: 57174 NXDomain* 0/0/0 (44)
08:22:37.142567 IP (tos 0x0, ttl 64, id 16310, offset 0, flags [DF], proto UDP (17), length 71)
    192.168.0.206.60278 > compalhub.home.domain: 43093+ PTR? 87.0.168.192.in-addr.arpa. (43)
08:22:37.147150 IP (tos 0x0, ttl 64, id 10871, offset 0, flags [DF], proto UDP (17), length 71)
    compalhub.home.domain > 192.168.0.206.60278: 43093 NXDomain* 0/0/0 (43)
08:22:37.242321 IP (tos 0x0, ttl 64, id 44666, offset 0, flags [DF], proto UDP (17), length 70)
    192.168.0.206.52259 > compalhub.home.domain: 14007+ PTR? 1.0.168.192.in-addr.arpa. (42)
08:22:37.248710 IP (tos 0x0, ttl 64, id 10872, offset 0, flags [DF], proto UDP (17), length 98)
    compalhub.home.domain > 192.168.0.206.52259: 14007* 1/0/0 1.0.168.192.in-addr.arpa. PTR compalh
ub.home. (70)
08:22:38.157885 IP (tos 0x0, ttl 63, id 7481, offset 0, flags [none], proto ICMP (1), length 84)
    192.168.0.87 > 192.168.0.206: ICMP echo request, id 37027, seq 4, length 64
08:22:38.157904 IP (tos 0x0, ttl 64, id 24351, offset 0, flags [none], proto ICMP (1), length 84)
    192.168.0.206 > 192.168.0.87: ICMP echo reply, id 37027, seq 4, length 64
08:22:39.182061 IP (tos 0x0, ttl 63, id 7482, offset 0, flags [none], proto ICMP (1), length 84)
    192.168.0.87 > 192.168.0.206: ICMP echo request, id 37027, seq 5, length 64
08:22:39.182082 IP (tos 0x0, ttl 64, id 24503, offset 0, flags [none], proto ICMP (1), length 84)
    192.168.0.206 > 192.168.0.87: ICMP echo reply, id 37027, seq 5, length 64
^C
12 packets captured
12 packets received by filter
0 packets dropped by kernel
```


Pingujemy DG

```
(kali㉿kali)-[~]
$ sudo tcpdump -i eth0 -v host 192.168.0.1
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
08:38:20.874554 IP (tos 0x0, ttl 64, id 29768, offset 0, flags [DF], proto ICMP (1), length 84)
    192.168.0.206 > compalhub.home: ICMP echo request, id 36671, seq 1, length 64
08:38:20.877059 IP (tos 0x0, ttl 64, id 10886, offset 0, flags [none], proto ICMP (1), length 84)
    compalhub.home > 192.168.0.206: ICMP echo reply, id 36671, seq 1, length 64
08:38:20.968985 IP (tos 0x0, ttl 64, id 63933, offset 0, flags [DF], proto UDP (17), length 70)
    192.168.0.206.59813 > compalhub.home.domain: 63273+ PTR? 1.0.168.192.in-addr.arpa. (42)
08:38:20.974572 IP (tos 0x0, ttl 64, id 10887, offset 0, flags [DF], proto UDP (17), length 98)
    compalhub.home.domain > 192.168.0.206.59813: 63273* 1/0/0 1.0.168.192.in-addr.arpa. PTR compalh
ub.home. (70)
08:38:20.974625 IP (tos 0x0, ttl 64, id 59384, offset 0, flags [DF], proto UDP (17), length 72)
    192.168.0.206.37103 > compalhub.home.domain: 2778+ PTR? 206.0.168.192.in-addr.arpa. (44)
08:38:20.980278 IP (tos 0x0, ttl 64, id 10888, offset 0, flags [DF], proto UDP (17), length 72)
    compalhub.home.domain > 192.168.0.206.37103: 2778 NXDomain* 0/0/0 (44)
08:38:21.876112 IP (tos 0x0, ttl 64, id 29791, offset 0, flags [DF], proto ICMP (1), length 84)
    192.168.0.206 > compalhub.home: ICMP echo request, id 36671, seq 2, length 64
08:38:21.877590 IP (tos 0x0, ttl 64, id 10889, offset 0, flags [none], proto ICMP (1), length 84)
    compalhub.home > 192.168.0.206: ICMP echo reply, id 36671, seq 2, length 64
08:38:22.877645 IP (tos 0x0, ttl 64, id 29975, offset 0, flags [DF], proto ICMP (1), length 84)
    192.168.0.206 > compalhub.home: ICMP echo request, id 36671, seq 3, length 64
08:38:22.879664 IP (tos 0x0, ttl 64, id 10890, offset 0, flags [none], proto ICMP (1), length 84)
    compalhub.home > 192.168.0.206: ICMP echo reply, id 36671, seq 3, length 64
^C
10 packets captured
10 packets received by filter
0 packets dropped by kernel
```

Widzimy wychodzące
żądanie ICMP i
odpowiedź na nie od
bramy domyślnej
(compalhub.home).

Dzięki zastosowaniu
filtrowania nie
widzimy wymiany
pakietów z żadnym
innym hostem w sieci.

Brama jako src

Przy samym tylko odświeżeniu strony agh dostajemy wiele pakietów odpowiadających filtrowi. Widzimy z ilu dodatkowych usług (analityka, czcionki, zamieszczone filmy) korzysta przykładowa strona.

```
(kali@kali)-[~]
$ sudo tcpdump -i eth0 -v dst 192.168.0.206 and src 192.168.0.1
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
09:21:22.240107 IP (tos 0x0, ttl 64, id 11060, offset 0, flags [DF], proto UDP (17), length 76)
    compalhub.home.domain > 192.168.0.206.53163: 8361 1/0/0 www.agh.edu.pl. A 149.156.96.150 (48)
09:21:22.241218 IP (tos 0x0, ttl 64, id 11061, offset 0, flags [DF], proto UDP (17), length 88)
    compalhub.home.domain > 192.168.0.206.53163: 43690 1/0/0 www.agh.edu.pl. AAAA 2001:6d8:10:118c::6096 (60)
09:21:22.307171 IP (tos 0x0, ttl 64, id 11062, offset 0, flags [DF], proto UDP (17), length 72)
    compalhub.home.domain > 192.168.0.206.39118: 1911 NXDomain* 0/0/0 (44)
09:21:22.310585 IP (tos 0x0, ttl 64, id 11063, offset 0, flags [DF], proto UDP (17), length 98)
    compalhub.home.domain > 192.168.0.206.47817: 57529* 1/0/0 1.0.168.192.in-addr.arpa. PTR compalhub.home. (70)
09:21:22.522540 IP (tos 0x0, ttl 64, id 11064, offset 0, flags [DF], proto UDP (17), length 86)
    compalhub.home.domain > 192.168.0.206.35332: 20674 1/0/0 www.googletagmanager.com. A 142.250.186.200 (58)
09:21:22.523574 IP (tos 0x0, ttl 64, id 11065, offset 0, flags [DF], proto UDP (17), length 98)
    compalhub.home.domain > 192.168.0.206.35332: 38081 1/0/0 www.googletagmanager.com. AAAA 2a00:1450:401b:80d::2008 (70)
09:21:22.843294 IP (tos 0x0, ttl 64, id 11066, offset 0, flags [DF], proto UDP (17), length 82)
    compalhub.home.domain > 192.168.0.206.34514: 22777 1/0/0 fonts.googleapis.com. A 142.250.203.138 (54)
09:21:22.846222 IP (tos 0x0, ttl 64, id 11067, offset 0, flags [DF], proto UDP (17), length 94)
    compalhub.home.domain > 192.168.0.206.34514: 51711 1/0/0 fonts.googleapis.com. AAAA 2a00:1450:401b:80e::200a (66)
09:21:22.904244 IP (tos 0x0, ttl 64, id 11068, offset 0, flags [DF], proto UDP (17), length 79)
    compalhub.home.domain > 192.168.0.206.38773: 2586 1/0/0 fonts.gstatic.com. A 142.250.186.195 (51)
09:21:22.933865 IP (tos 0x0, ttl 64, id 11069, offset 0, flags [DF], proto UDP (17), length 91)
    compalhub.home.domain > 192.168.0.206.38773: 40472 1/0/0 fonts.gstatic.com. AAAA 2a00:1450:401b:804::2003 (63)
09:21:22.946375 IP (tos 0x0, ttl 64, id 11070, offset 0, flags [DF], proto UDP (17), length 202)
    compalhub.home.domain > 192.168.0.206.41605: 16290 8/0/0 img.youtube.com. CNAME ytimg.l.google.com., ytimg.l.google.co
m. A 216.58.215.78, ytimg.l.google.com. A 142.250.203.206, ytimg.l.google.com. A 142.250.203.142, ytimg.l.google.com. A 17
2.217.16.46, ytimg.l.google.com. A 216.58.209.14, ytimg.l.google.com. A 216.58.215.110, ytimg.l.google.com. A 142.250.186.
206 (174)
09:21:22.959655 IP (tos 0x0, ttl 64, id 11071, offset 0, flags [DF], proto UDP (17), length 202)
    compalhub.home.domain > 192.168.0.206.41605: 47008 5/0/0 img.youtube.com. CNAME ytimg.l.google.com., ytimg.l.google.co
m. AAAA 2a00:1450:401b:807::200e, ytimg.l.google.com. AAAA 2a00:1450:401b:805::200e, ytimg.l.google.com. AAAA 2a00:1450:40
1b:806::200e, ytimg.l.google.com. AAAA 2a00:1450:401b:808::200e (174)
09:21:22.963505 IP (tos 0x0, ttl 64, id 11072, offset 0, flags [DF], proto UDP (17), length 71)
    compalhub.home.domain > 192.168.0.206.55275: 56918 1/0/0 ipinfo.io. A 34.117.59.81 (43)
09:21:22.975784 IP (tos 0x0, ttl 64, id 11073, offset 0, flags [DF], proto UDP (17), length 148)
    compalhub.home.domain > 192.168.0.206.55275: 56919 0/1/0 (120)
09:21:23.616193 ARP, Ethernet (len 6), IPv4 (len 4), Reply compalhub.home is-at 38:43:7d:65:6c:72 (oui Unknown), length 46
^C
15 packets captured
15 packets received by filter
0 packets dropped by kernel
```

Nasłuchujemy na połączenia do portu 80.

Port 80. obsługuje
protokół HTTP, a
więc protokół
nieszyfrowany.

Możemy podejrzeć
treść przesyłanych
żądań HTTP.

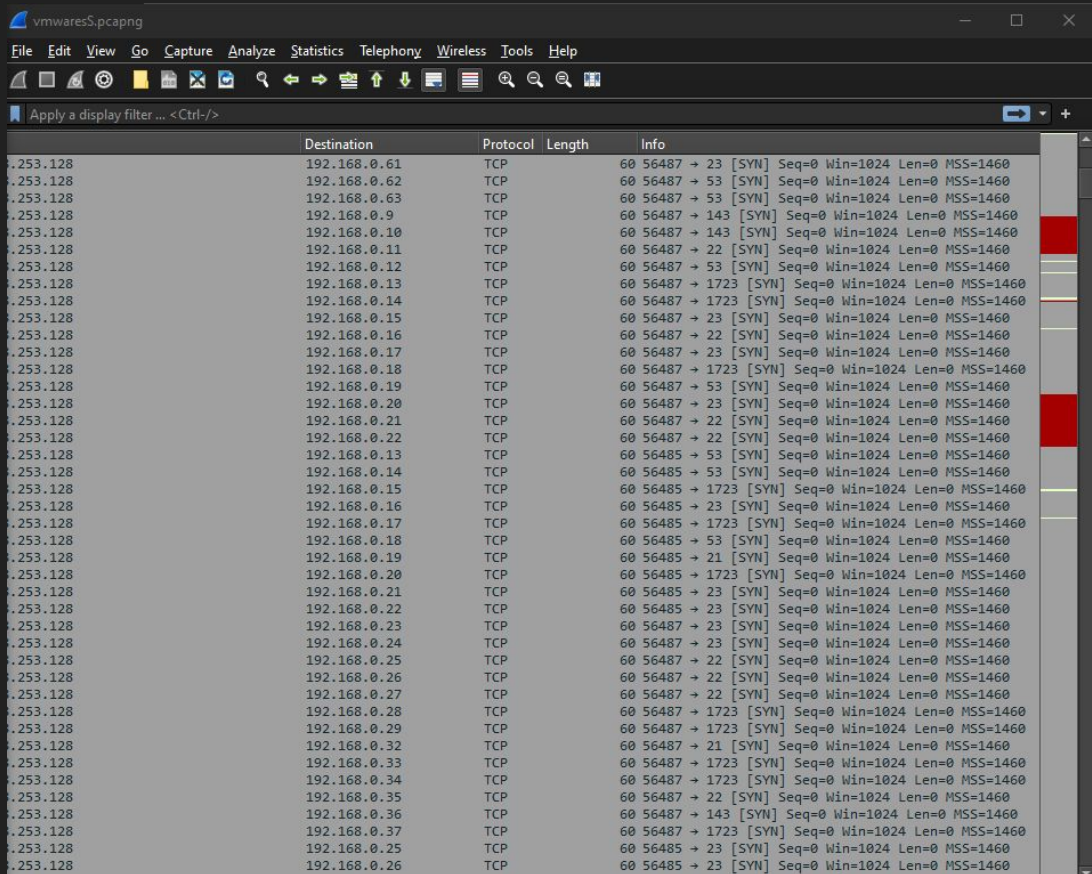
```
(kali@kali)-[~]  
$ sudo tcpdump -i eth0 -v port 80  
  
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
09:36:37.817206 IP (tos 0x0, ttl 64, id 45817, offset 0, flags [DF], proto TCP (6), length 60)  
    192.168.0.206.55842 > myshopify.com.http: Flags [S], cksum 0xffc8 (incorrect → 0xf93b), seq 3210156857, win 32120,  
    options [mss 1460,sackOK,TS val 867428602 ecr 0,nop,wscale 7], length 0  
09:36:37.817232 IP (tos 0x0, ttl 64, id 11770, offset 0, flags [DF], proto TCP (6), length 60)  
    192.168.0.206.55844 > myshopify.com.http: Flags [S], cksum 0xffc8 (incorrect → 0xadb4), seq 3091885515, win 32120,  
    options [mss 1460,sackOK,TS val 867428602 ecr 0,nop,wscale 7], length 0  
09:36:37.841344 IP (tos 0x0, ttl 57, id 0, offset 0, flags [DF], proto TCP (6), length 60)  
    myshopify.com.http > 192.168.0.206.55842: Flags [S.], cksum 0x4e6c (correct), seq 916467781, ack 3210156858, win 655  
    35, options [mss 1420,sackOK,TS val 1590715043 ecr 867428602,nop,wscale 13], length 0  
09:36:37.841370 IP (tos 0x0, ttl 64, id 45818, offset 0, flags [DF], proto TCP (6), length 52)  
    192.168.0.206.55842 > myshopify.com.http: Flags [.], cksum 0xffc0 (incorrect → 0x7c02), ack 1, win 251, options [no  
    p,nop,TS val 867428627 ecr 1590715043], length 0  
09:36:37.841716 IP (tos 0x0, ttl 64, id 45819, offset 0, flags [DF], proto TCP (6), length 391)  
    192.168.0.206.55842 > myshopify.com.http: Flags [P.], cksum 0x0114 (incorrect → 0xb916), seq 1:340, ack 1, win 251,  
    options [nop,nop,TS val 867428627 ecr 1590715043], length 339: HTTP, length: 339  
    GET / HTTP/1.1  
    Host: swagoffroad.com  
    User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/115.0  
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8  
    Accept-Language: en-US,en;q=0.5  
    Accept-Encoding: gzip, deflate  
    Connection: keep-alive  
    Upgrade-Insecure-Requests: 1  
  
09:36:37.846174 IP (tos 0x0, ttl 57, id 0, offset 0, flags [DF], proto TCP (6), length 60)  
    myshopify.com.http > 192.168.0.206.55844: Flags [S.], cksum 0xe4dd (correct), seq 1425718987, ack 3091885516, win 65  
    535, options [mss 1420,sackOK,TS val 787874724 ecr 867428602,nop,wscale 13], length 0  
09:36:37.846188 IP (tos 0x0, ttl 64, id 11771, offset 0, flags [DF], proto TCP (6), length 52)  
    192.168.0.206.55844 > myshopify.com.http: Flags [.], cksum 0xffc0 (incorrect → 0x1270), ack 1, win 251, options [no  
    p,nop,TS val 867428631 ecr 787874724], length 0
```


Nasłuchujemy połączenia do portu 443.

Port 443. obsługuje połączenia za pomocą HTTPS (HTTP Secure), który jest protokołem szyfrowanym. Wszystko, co możemy podejrzeć za pomocą tcpdump, to informacje znajdujące się na poniższym zrzucie ekranu (wybrałem tylko trzy pakiety - w rzeczywistości było ich o wiele więcej, lecz wszystkie były podobne w swej strukturze).

```
(kali@kali)-[~]  
$ sudo tcpdump -i eth0 -v port 443  
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
09:34:31.400250 IP6 (flowlabel 0xb34cb, hlim 64, next-header UDP (17) payload length: 699) 2a02:a31a:e08f:4600:ef84:ae83:98a7:9ee6.44312 > waw02s16-in-x04.1e100.net.https: [bad udp cksum 0x7285 → 0x260a!] UDP, length 691  
09:34:31.436048 IP6 (hlim 58, next-header UDP (17) payload length: 43) waw02s16-in-x04.1e100.net.https > 2a02:a31a:e08f:4600:ef84:ae83:98a7:9ee6.44312: [udp sum ok] UDP, length 35  
09:34:31.457378 IP6 (flowlabel 0xb34cb, hlim 64, next-header UDP (17) payload length: 42) 2a02:a31a:e08f:4600:ef84:ae83:98a7:9ee6.44312 > waw02s16-in-x04.1e100.net.https: [bad udp cksum 0x6ff4 → 0x7d0c!] UDP, length 34
```


Skanujemy z użyciem sudo nmap -sS 192.168.0.0/24



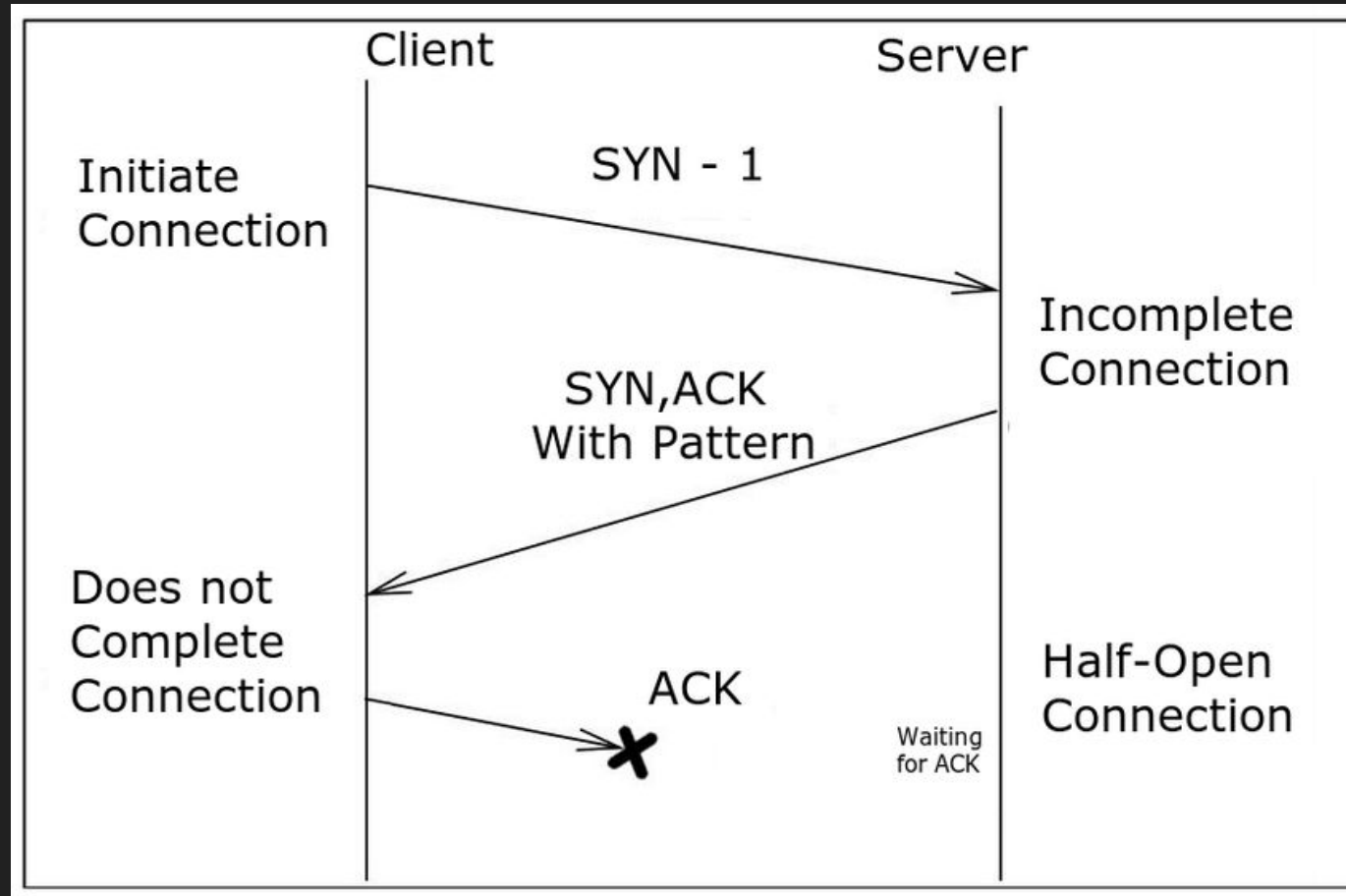
	Destination	Protocol	Length	Info
1.253.128	192.168.0.61	TCP	60	56487 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2.253.128	192.168.0.62	TCP	60	56487 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
3.253.128	192.168.0.63	TCP	60	56487 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
4.253.128	192.168.0.9	TCP	60	56487 → 143 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
5.253.128	192.168.0.10	TCP	60	56487 → 143 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.253.128	192.168.0.11	TCP	60	56487 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
7.253.128	192.168.0.12	TCP	60	56487 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
8.253.128	192.168.0.13	TCP	60	56487 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
9.253.128	192.168.0.14	TCP	60	56487 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
10.253.128	192.168.0.15	TCP	60	56487 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
11.253.128	192.168.0.16	TCP	60	56487 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
12.253.128	192.168.0.17	TCP	60	56487 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
13.253.128	192.168.0.18	TCP	60	56487 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
14.253.128	192.168.0.19	TCP	60	56487 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
15.253.128	192.168.0.20	TCP	60	56487 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
16.253.128	192.168.0.21	TCP	60	56487 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
17.253.128	192.168.0.22	TCP	60	56487 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
18.253.128	192.168.0.13	TCP	60	56485 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
19.253.128	192.168.0.14	TCP	60	56485 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
20.253.128	192.168.0.15	TCP	60	56485 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
21.253.128	192.168.0.16	TCP	60	56485 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
22.253.128	192.168.0.17	TCP	60	56485 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
23.253.128	192.168.0.18	TCP	60	56485 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
24.253.128	192.168.0.19	TCP	60	56485 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
25.253.128	192.168.0.20	TCP	60	56485 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
26.253.128	192.168.0.21	TCP	60	56485 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
27.253.128	192.168.0.22	TCP	60	56485 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
28.253.128	192.168.0.23	TCP	60	56487 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
29.253.128	192.168.0.24	TCP	60	56487 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
30.253.128	192.168.0.25	TCP	60	56487 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
31.253.128	192.168.0.26	TCP	60	56487 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
32.253.128	192.168.0.27	TCP	60	56487 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
33.253.128	192.168.0.28	TCP	60	56487 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
34.253.128	192.168.0.29	TCP	60	56487 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
35.253.128	192.168.0.32	TCP	60	56487 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
36.253.128	192.168.0.33	TCP	60	56487 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
37.253.128	192.168.0.34	TCP	60	56487 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
38.253.128	192.168.0.35	TCP	60	56487 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
39.253.128	192.168.0.36	TCP	60	56487 → 143 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
40.253.128	192.168.0.37	TCP	60	56487 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
41.253.128	192.168.0.25	TCP	60	56485 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
42.253.128	192.168.0.26	TCP	60	56485 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

```
(kali㉿kali)-[~]  
$ sudo nmap -sS 192.168.0.0/24
```

Podanie maski /24 jest konieczne aby przeskanować całą podsieć.

Jest to skanowanie typu TCP half-open, o czym świadczy bardzo duża ilość komunikatów SYN wysyłanych przez atakującą maszynę i brak komunikatów ACK z jej strony.

Takie skanowanie przeprowadza się licząc na to, że niepełne połączenia TCP nie będą odnotowane w logach systemów ofiary. Jednak przy odpowiedniej konfiguracji da się zarejestrować i takie skanowanie



Skanujemy z użyciem sudo nmap 192.168.0.0/24 -data-length 32 -f -T5

vmware13.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
5808.	2024-12-18 19:18:34.954204	192.168.253.128	192.168.0.47	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=0, ID=326c) [Reassembled in #580818]
5808.	2024-12-18 19:18:34.954227	192.168.253.128	192.168.0.47	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=8, ID=326c) [Reassembled in #580818]
5808.	2024-12-18 19:18:34.954229	192.168.253.128	192.168.0.47	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=16, ID=326c) [Reassembled in #580818]
5808.	2024-12-18 19:18:34.954254	192.168.253.128	192.168.0.47	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=24, ID=326c) [Reassembled in #580818]
5808.	2024-12-18 19:18:34.954256	192.168.253.128	192.168.0.47	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=32, ID=326c) [Reassembled in #580818]
5808.	2024-12-18 19:18:34.954278	192.168.253.128	192.168.0.47	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=40, ID=326c) [Reassembled in #580818]
5808.	2024-12-18 19:18:34.954281	192.168.253.128	192.168.0.47	TCP	60	45075 → 32781 [SYN] Seq=0 Win=1024 Len=32 MSS=1460
5808.	2024-12-18 19:18:34.954307	192.168.253.128	192.168.0.48	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=0, ID=5992) [Reassembled in #580825]
5808.	2024-12-18 19:18:34.954309	192.168.253.128	192.168.0.48	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=8, ID=5992) [Reassembled in #580825]
5808.	2024-12-18 19:18:34.954331	192.168.253.128	192.168.0.48	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=16, ID=5992) [Reassembled in #580825]
5808.	2024-12-18 19:18:34.954334	192.168.253.128	192.168.0.48	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=24, ID=5992) [Reassembled in #580825]
5808.	2024-12-18 19:18:34.954358	192.168.253.128	192.168.0.48	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=32, ID=5992) [Reassembled in #580825]
5808.	2024-12-18 19:18:34.954360	192.168.253.128	192.168.0.48	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=40, ID=5992) [Reassembled in #580825]
5808.	2024-12-18 19:18:34.954395	192.168.253.128	192.168.0.48	TCP	60	45075 → 1152 [SYN] Seq=0 Win=1024 Len=32 MSS=1460
5808.	2024-12-18 19:18:34.954387	192.168.253.128	192.168.0.49	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=0, ID=2e6e) [Reassembled in #580832]
5808.	2024-12-18 19:18:34.954412	192.168.253.128	192.168.0.49	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=8, ID=2e6e) [Reassembled in #580832]
5808.	2024-12-18 19:18:34.954413	192.168.253.128	192.168.0.49	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=16, ID=2e6e) [Reassembled in #580832]
5808.	2024-12-18 19:18:34.954436	192.168.253.128	192.168.0.49	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=24, ID=2e6e) [Reassembled in #580832]
5808.	2024-12-18 19:18:34.954438	192.168.253.128	192.168.0.49	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=32, ID=2e6e) [Reassembled in #580832]
5808.	2024-12-18 19:18:34.954462	192.168.253.128	192.168.0.49	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=40, ID=2e6e) [Reassembled in #580832]
5808.	2024-12-18 19:18:34.954464	192.168.253.128	192.168.0.49	TCP	60	45075 → 4848 [SYN] Seq=0 Win=1024 Len=32 MSS=1460
5808.	2024-12-18 19:18:34.954509	192.168.253.128	192.168.0.50	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=0, ID=840f) [Reassembled in #580839]
5808.	2024-12-18 19:18:34.954512	192.168.253.128	192.168.0.50	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=8, ID=840f) [Reassembled in #580839]
5808.	2024-12-18 19:18:34.954535	192.168.253.128	192.168.0.50	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=16, ID=840f) [Reassembled in #580839]
5808.	2024-12-18 19:18:34.954537	192.168.253.128	192.168.0.50	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=24, ID=840f) [Reassembled in #580839]
5808.	2024-12-18 19:18:34.954558	192.168.253.128	192.168.0.50	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=32, ID=840f) [Reassembled in #580839]
5808.	2024-12-18 19:18:34.954560	192.168.253.128	192.168.0.50	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=40, ID=840f) [Reassembled in #580839]
5808.	2024-12-18 19:18:34.954583	192.168.253.128	192.168.0.50	TCP	60	45075 → 5226 [SYN] Seq=0 Win=1024 Len=32 MSS=1460
5808.	2024-12-18 19:18:34.954585	192.168.253.128	192.168.0.51	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=0, ID=9f3e) [Reassembled in #580846]
5808.	2024-12-18 19:18:34.954606	192.168.253.128	192.168.0.51	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=8, ID=9f3e) [Reassembled in #580846]
5808.	2024-12-18 19:18:34.954607	192.168.253.128	192.168.0.51	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=16, ID=9f3e) [Reassembled in #580846]
5808.	2024-12-18 19:18:34.954628	192.168.253.128	192.168.0.51	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=24, ID=9f3e) [Reassembled in #580846]
5808.	2024-12-18 19:18:34.954630	192.168.253.128	192.168.0.51	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=32, ID=9f3e) [Reassembled in #580846]
5808.	2024-12-18 19:18:34.954652	192.168.253.128	192.168.0.51	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=40, ID=9f3e) [Reassembled in #580846]
5808.	2024-12-18 19:18:34.954654	192.168.253.128	192.168.0.51	TCP	60	45075 → 32781 [SYN] Seq=0 Win=1024 Len=32 MSS=1460
5808.	2024-12-18 19:18:34.954709	192.168.253.128	192.168.0.15	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=0, ID=bd70) [Reassembled in #580853]
5808.	2024-12-18 19:18:34.954711	192.168.253.128	192.168.0.15	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=8, ID=bd70) [Reassembled in #580853]
5808.	2024-12-18 19:18:34.954733	192.168.253.128	192.168.0.15	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=16, ID=bd70) [Reassembled in #580853]
5808.	2024-12-18 19:18:34.954734	192.168.253.128	192.168.0.15	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=24, ID=bd70) [Reassembled in #580853]
5808.	2024-12-18 19:18:34.954735	192.168.253.128	192.168.0.15	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=32, ID=bd70) [Reassembled in #580853]
5808.	2024-12-18 19:18:34.954757	192.168.253.128	192.168.0.15	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=40, ID=bd70) [Reassembled in #580853]
5808.	2024-12-18 19:18:34.954780	192.168.253.128	192.168.0.15	TCP	60	45073 → 1152 [SYN] Seq=0 Win=1024 Len=32 MSS=1460
5808.	2024-12-18 19:18:34.954782	192.168.253.128	192.168.0.16	IPv4	60	Fragmented IP protocol (proto=TCP 6, off=0, ID=6c9f) [Reassembled in #580860]

Jest to cichszy tryb skanowania, w którym fragmentujemy pakiety i do każdego z nich dodajemy 32 losowe bajty, aby utrudnić skanowania. Mniej rzuca się ono w oczy od poprzedniego, jednak

nie jest to znacząca różnica. Moglibyśmy spróbować poprawić jego dyskretność poprzez zmianę odstępu czasu pomiędzy poszczególnymi pakietami (opcja T).

Zad 5. Analiza pliku .pcap

- a) Adres IP komputera poddanego analizie: 172.16.17.131
- b) Najpewniej jest to 172.16.17.2, ponieważ z nim najczęściej zachodzi komunikacja (nie licząc skanowania).
- c) Tak. Po kliknięciu na pakiet widzimy:

```
Frame 27: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0  
Ethernet II, Src: VMware_ec:8a:14 (00:0c:29:ec:8a:14), Dst: VMware_24:d0:a3 (00:0c:29:24:d0:a3)  
Internet Protocol Version 4, Src: 172.16.17.128, Dst: 172.16.17.131  
Transmission Control Protocol, Src Port: 57324, Dst Port: 21, Seq: 0, Len: 0
```

Ponadto ich MAC (część określająca wytwórcę) są zgodne z adresami przypisanymi VMware (00:0c:29).

- d) Tak.
- e) Była skanowana metodą TCP half-open (SYN scan) przez adres IP 172.16.17.128. Można to też określić mianem ataku DOS typu SYN Flood (otwieranie wielu połączeń i zajmowanie zasobów bez zamiaru faktycznego z nich skorzystania).

Zad 5. Analiza pliku .pcap cd.

f) Należy on do IP 172.16.17.128, czyli do atakującego.

g) Jest to plik bad_file.exe i został pobrany z 172.16.17.128 o 18:37 (ukończenie pobierania).

h)

```
PS C:\Users\barte> Get-FileHash "D:\Users\barte\Downloads\%5cbad_file.exe" -Algorithm MD5
```

Algorithm	Hash	Path
-----	----	----
MD5	C654E9EB9B0528CA65877930A44B601A	D:\Users\barte\Downloads\%5cb..

i) Po wyszukaniu hashu w VirusTotalu możemy być pewni, że analizowany plik jest złośliwym oprogramowaniem. Znaczna większość z silników antywirusowych potwierdza jego szkodliwość i oznacza jako wirusa.

Wyniki wyszukiwania hasha w VirusTotalu

Wynik nie
pozostawia
wątpliwości co do
tego, czym mamy
tutaj do czynienia.

51
/ 68

Community
Score

51/68 security vendors flagged this file as malicious

133c762b3f0806e0f07bd66b28fa2c79f3265478189b86b95349d0aad323d9c

ab.exe

Size
72.07 KB

Last Analysis Date
3 years ago

EXE

peexe

overlay

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 4

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis

Do you want to automate checks?

Ad-Aware	Trojan.CryptZ.Gen	AhnLab-V3	Trojan/Win32.Shell.R1283
ALYac	Trojan.CryptZ.Gen	Arcabit	Trojan.CryptZ.Gen
Avast	Win32:SwPatch [Wrm]	AVG	Win32:SwPatch [Wrm]
Avira (no cloud)	TR/Patched.Gen2	BitDefender	Trojan.CryptZ.Gen
BitDefenderTheta	Gen:NN.ZexaF.34062.eq1@aGwt0wfi	Bkav Pro	W32.FamVT.RorenNHC.Trojan
ClamAV	Win.Trojan.Swrort-5710536-0	Comodo	TrojWare.Win32.Rozena.A@4jwdqr
CrowdStrike Falcon	Win/malicious_confidence_100% (D)	Cybereason	Malicious.b9b052
Cylance	Unsafe	Cynet	Malicious (score: 100)
Cyren	W32/Swrort.D	eGambit	Unsafe.AI_Score_97%
Elastic	Malicious (high confidence)	Emnara	Trojan.CryptZ.Gen (D)

Zad 5. Analiza pliku .pcap cd.

j) Były to porty 57324 oraz 57325 po stronie atakującego. Po stronie ofiary - bardzo różne.

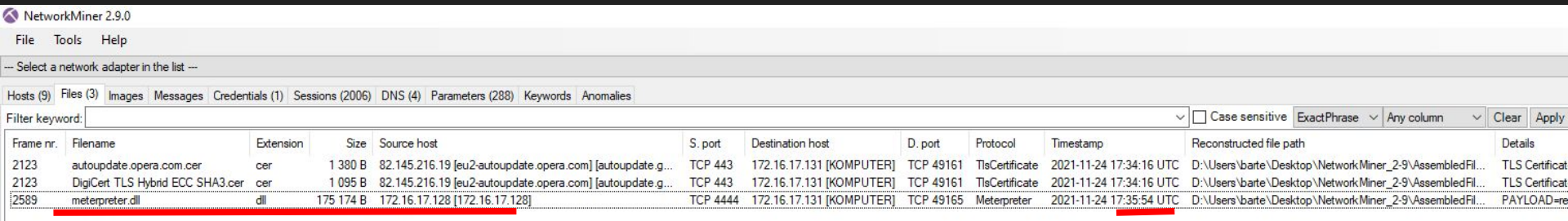
k) Nazwa tego komputera to KOMPUTER.

```
Microsoft Windows Browser Protocol
  Command: Host Announcement (0x01)
  Update Count: 0
  Update Periodicity: 4 minutes
  Host Name: KOMPUTER
  Windows version: Windows 7 or Windows Server 2008 R2
  OS Major Version: 6
  OS Minor Version: 1
  Server Type: 0x00001003, Workstation, Server, NT Workstation
    ....1 = Workstation: This is a Workstation
    ...1. = Server: This is a Server
    ...0.. = SQL: This is NOT an SQL server
    ...0... = Domain Controller: This is NOT a Domain Controller
    ...0 .... = Backup Controller: This is NOT a Backup Controller
    ...0 ..... = Time Source: This is NOT a Time Source
```

Zad 6. Użyjmy Network Minera

Czy w trakcie wykorzystania programu NetworkMiner jesteśmy w stanie odszukać zainfekowany plik? Jeśli nie, to dlaczego, jeśli tak to w jaki sposób?

Tak, możemy. Musimy wejść najpierw w zakładkę pliki. Wygląda ona tak:



NetworkMiner 2.9.0

File Tools Help

--- Select a network adapter in the list ---

Hosts (9) Files (3) Images Messages Credentials (1) Sessions (2006) DNS (4) Parameters (288) Keywords Anomalies

Filter keyword: ☐ Case sensitive ExactPhrase Any column Clear Apply

Frame nr.	Filename	Extension	Size	Source host	S. port	Destination host	D. port	Protocol	Timestamp	Reconstructed file path	Details
2123	autoupdate.opera.com.cer	cer	1 380 B	82.145.216.19 [eu2-autoupdate.opera.com] [autoupdate.g...	TCP 443	172.16.17.131 [KOMPUTER]	TCP 49161	TlsCertificate	2021-11-24 17:34:16 UTC	D:\Users\barte\Desktop\NetworkMiner_2-9\AssembledFil...	TLS Certificat
2123	DigiCert TLS Hybrid ECC SHA3.cer	cer	1 095 B	82.145.216.19 [eu2-autoupdate.opera.com] [autoupdate.g...	TCP 443	172.16.17.131 [KOMPUTER]	TCP 49161	TlsCertificate	2021-11-24 17:34:16 UTC	D:\Users\barte\Desktop\NetworkMiner_2-9\AssembledFil...	TLS Certificat
2589	meterpreter.dll	dll	175 174 B	172.16.17.128 [172.16.17.128]	TCP 4444	172.16.17.131 [KOMPUTER]	TCP 49165	Meterpreter	2021-11-24 17:35:54 UTC	D:\Users\barte\Desktop\NetworkMiner_2-9\AssembledFil...	PAYLOAD=...

Meterpreter to payload w ataku z wykorzystaniem narzędzia Metasploit, który zapewnia interaktywną powłokę. Mamy czas jego pobrania. Sprawdźmy ten moment w zakładce Parameters.

Mamy nazwę złośliwego pliku!

NetworkMiner 2.9.0

File Tools Help

-- Select a network adapter in the list --

Hosts (9) Files (3) Images Messages Credentials (1) Sessions (2006) DNS (4) Parameters (288) Keywords Anomalies

Filter keyword: ☐ Case sensitive ExactPhrase Any column Clear Apply

Parameter name	Parameter value	Frame number	Source host	Source port	Destination host	Destination port	Timestamp
SMB NT Create AndX Request 2304	\\bad_file.exe	2277	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:52 U
SMB NT Create AndX Request 2880	\\bad_file.exe	2298	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:52 U
SMB NT Create AndX Request 3072	\\bad_file.exe.Manifest	2307	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:52 U
SMB NT Create AndX Request 3200	\\bad_file.exe	2311	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:52 U
SMB NT Create AndX Request 3264	\\bad_file.exe	2313	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:52 U
SMB NT Create AndX Request 3904	\\bad_file.exe	2349	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:52 U
SMB NT Create AndX Request 4672	\\bad_file.exe	2400	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:53 U
SMB NT Create AndX Request 5056	\\	2418	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:53 U
SMB NT Create AndX Request 5504	\\bad_file.exe	2432	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:53 U
SMB NT Create AndX Request 6016	\\bad_file.exe.Manifest	2451	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:53 U
SMB NT Create AndX Request 6144	\\bad_file.exe	2455	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:53 U
SMB NT Create AndX Request 6208	\\bad_file.exe	2457	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:53 U
SMB NT Create AndX Request 6656	\\bad_file.exe	2484	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:54 U
SMB NT Create AndX Request 6784	\\bad_file.exe	2488	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:54 U
SMB NT Create AndX Request 7168	\\bad_file.exe	2500	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:54 U
SMB NT Create AndX Request 7296	\\bad_file.exe	2504	172.16.17.131 [KOMPUTER]	TCP 49162	172.16.17.128 [172.16.17.128]	TCP 445	2021-11-24 17:35:54 U

Oczywiście, identyczną jak w Wiresharku.

Dziękuję za uwagę!